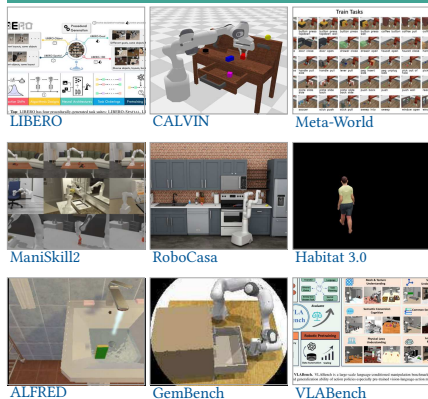


Predict or Act?

the benchmark landscape on one axis; each tile links to its paper

CLOSED-LOOP task-success suites



what does it
score?

← task success
(closed loop)

VS

prediction quality
(open loop) →

the survey's org-
anising axis

only 11 of 160
test if predic-
tion helps action

OPEN-LOOP world-model & video eval

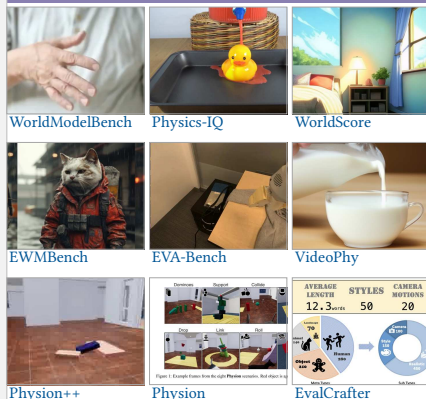


Fig. 1. **Predict or Act?** The survey's organising question, posed with real benchmarks. *Closed-loop* (left): policy and embodied suites score task success by running a policy in the environment. *Open-loop* (right): world-model and video-generation benchmarks score prediction or generation quality with no action. The centre axis is the question neither pole answers, whether prediction earns a closed-loop advantage, built explicitly by only 11 of 160 benchmarks (Table ??). Every tile is a representative figure from the benchmark's paper and links to it; per-tile provenance is in Table ?. Images © their respective authors, reproduced for scholarly review.